UNICITY PROTOCOL · STRATEGIC PROPOSAL · 2026

SECURING THE AGENTIC ECONOMY.

Proof of identity and permission, bound to the payment itself.

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PREPARED FOR
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ACT I · THE SHIFT

THE AGENTS ARE HERE. THE RAILS ARE NOT.

Most of the internet is already machines — and they have started to pay, at software speed.

57.5%

OF WEB REQUESTS ARE AUTOMATED, NOT HUMAN — CLOUDFLARE, 2026

Legacy networks cannot verify an agent's authority or hold it to a budget without a centralized chokepoint. For machines to transact, the **proof of authority has to travel with the money**.

ACT II · WHY THE RAILS FAIL

A MACHINE CANNOT WAIT FOR A SHARED LEDGER.

The 17-year-old design behind every blockchain is a permanent ceiling on scale and privacy.

To add one transaction, every node must broadcast, order, validate, and record it. Machine-speed commerce requires removing the **shared ledger** entirely.

BROADCAST

Every node hears every transaction.

ORDER

Global agreement on the sequence of all of them.

VALIDATE

Every node re-runs every one.

RECORD

Every node stores the whole state, forever.

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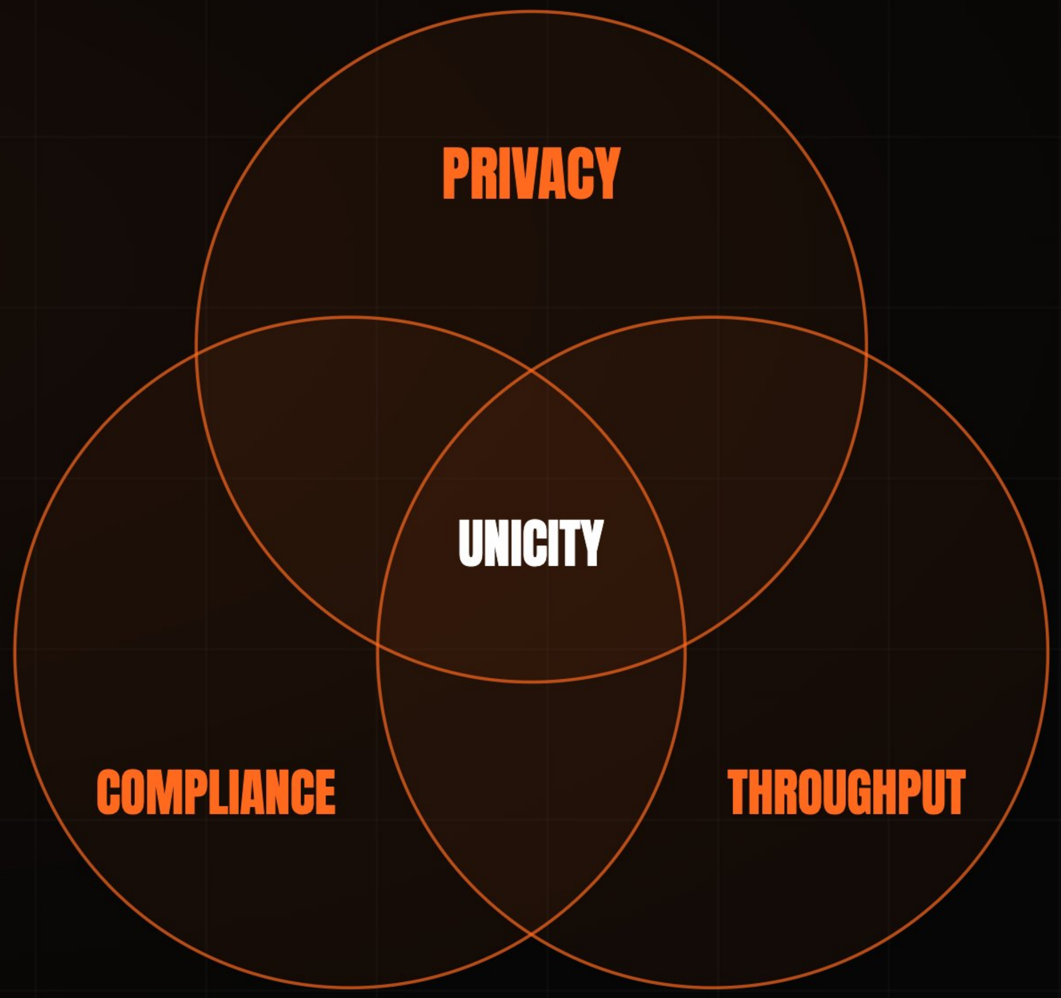
ACT II · WHY THE RAILS FAIL

EVERY DIGITAL DOLLAR GIVES ONE UP.

Compliance, privacy, throughput — every design so far has been forced to pick two.

The whole payment landscape is trapped in the trade-off. Unicity is the first to hold **all three at once** — because it never places the transaction on a shared ledger to begin with.



ACT II · THE GAP

FAIR ACCESS STILL LACKS ONE THING.

Disintermediation and digital value are solved. Proving who is on the other end was never built.

01	**DISINTERMEDIATION** Open infrastructure bypassed the legacy financial middleman.	SOLVED
02	**DIGITAL VALUE** Stablecoins now settle tens of trillions of dollars a year.	SOLVED
03	**CRYPTOGRAPHIC IDENTITY** No blockchain can natively verify a recipient's legal or operational standing before a transaction executes.	UNSOLVED

Unicity was engineered to close this final gap.

ACT II · THE GAP

WHICH IS WHY IDENTITY KEEPS FAILING.

Bolt-on verification sits beside the transaction — and a check that can be skipped gets skipped.

Self-sovereign identity stalled because the credential sat beside the transaction. Identity has to live **inside the asset** — enforceable across any jurisdiction, not optional at the application layer.

THE MARKET IS SPENDING BILLIONS – AND STILL MISSING IT.

Everyone is optimizing the ledger or splitting the proof across layers. No one keeps it whole.

That spend is the validation: the problem is real and unsolved. **Unicity keeps authorization, settlement, and the audit trail whole, inside the asset.**

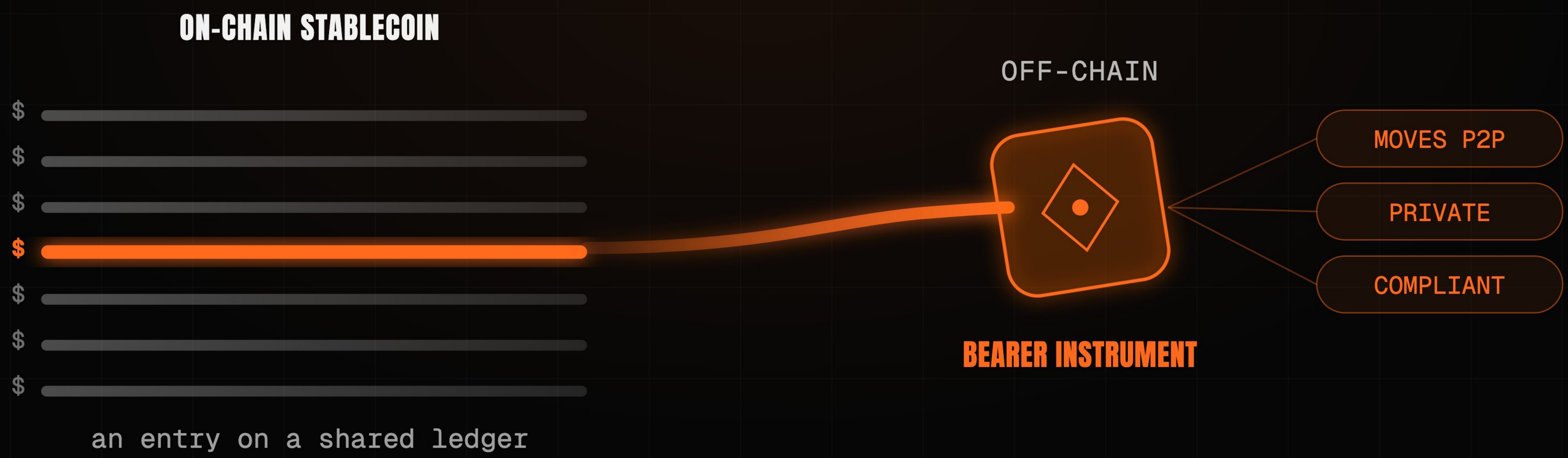
PLAYER	THEIR MOVE	APPROACH
CIRCLE	Pivoted to infrastructure — the Arc L1, a $222M presale, bought Malachite.	optimise the ledger
SOLANA	Alpenglow cut finality to ~150ms; its co-founder concedes a physical ceiling.	optimise the ledger
AP2 · X402	Authorize in one layer, settle in another — the proof splits across the stack.	split the proof
UNICITY	Keeps authorization, settlement, and the audit trail inside the asset.	keep it whole

ACT III · THE ANSWER

SO TAKE THE DOLLAR OFF THE LEDGER.

Off-chain settlement restores the properties of physical cash to a digital stablecoin.

Unicity converts an on-chain stablecoin into a **self-contained, self-proving bearer instrument**. It carries its own proof and moves **peer-to-peer** over any channel — HTTP, QR, NOSTR — with zero ledger lookups.



ACT III · THE ANSWER

SO THE NETWORK ATTESTS ONE THING.

Each generation of consensus removed work from the network. Unicity keeps only the irreducible function.

The **Uniqueness Oracle** answers one question: has this token been spent? It never re-executes transactions or reads payloads, so throughput scales horizontally — **30,000 tx/sec per shard, by design.**

CONSENSUS & EMISSION
decentralized base security

↓ certified state root

UNIQUENESS ORACLE
stores only proofs that token states are unique

↓ inclusion proofs

EXECUTION LAYER
agents and users transact client-side, off-chain

2009 · BITCOIN

CORRECTNESS + ORDERING

Every node certifies every transaction and agrees the global order.

2023 · SUI · FASTPAY

CORRECTNESS ONLY

Global ordering dropped for assets that share no state.

2026 · UNICITY

UNIQUENESS ONLY

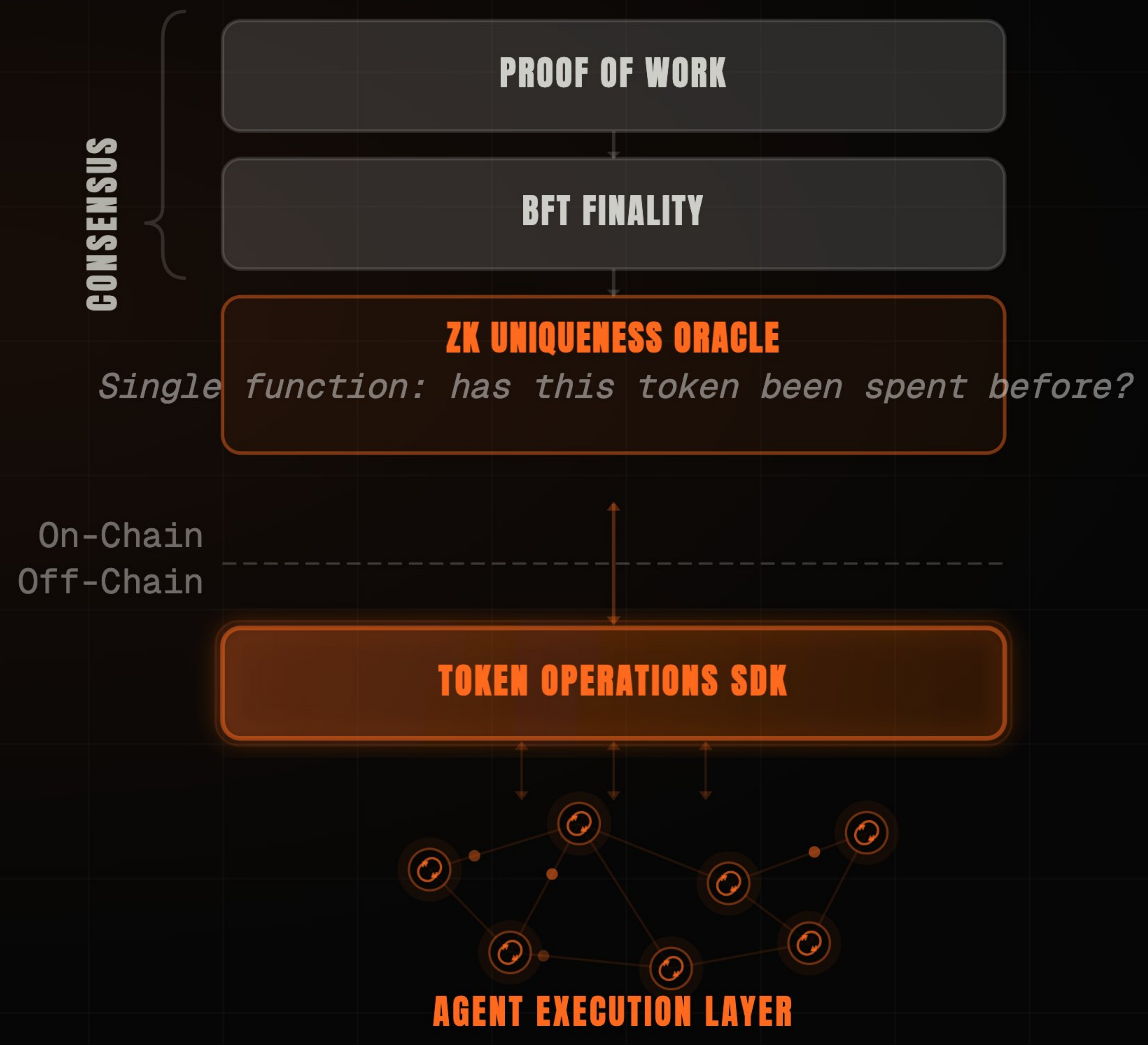
The network attests one thing. Correctness moves to the edge.

ACT III · THE ANSWER

SO THE CHAIN DOES ONE JOB.

Consensus stays minimal on-chain; everything an agent touches runs off-chain.

Proof-of-work, BFT finality, and the ZK oracle keep the chain minimal — the SDK and the agent execution layer run beside it. **The chain does one job; the economy runs off it.**



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ACT III · THE ANSWER

NOW COMPLIANCE NEEDS NO MIDDLEMAN.

Gate a transfer cryptographically, without surrendering peer-to-peer privacy.

The **Receive Predicate** programs KYC, jurisdiction, and accreditation directly into the asset. If the recipient cannot satisfy it locally, the transfer mathematically fails — **the asset enforces its own compliance**.

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ACT III · THE ANSWER

AND THE OPEN LEDGER GOES DARK.

Absolute confidentiality is the prerequisite for institutional capital flows.

Privacy is a founding property of the protocol, not an afterthought — mathematically proven against all three observers.

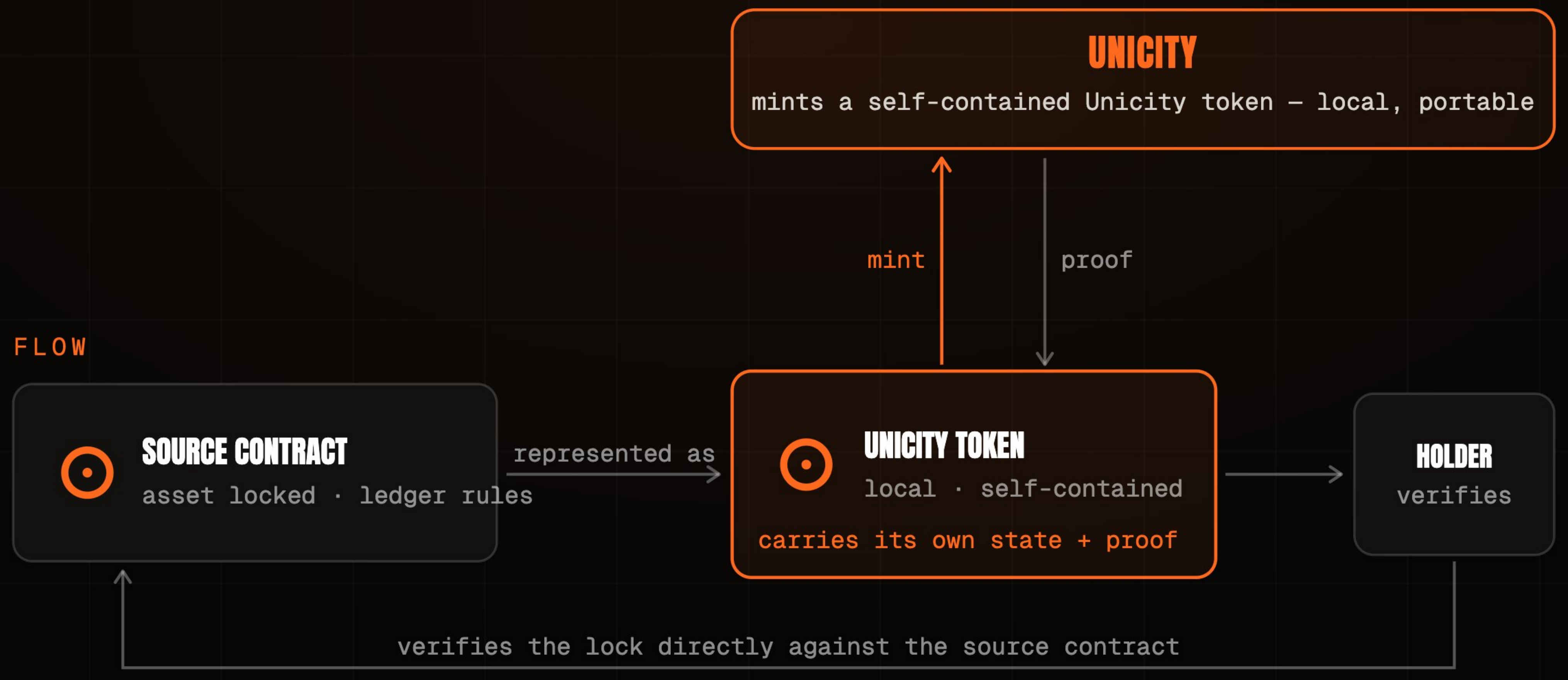
THE NETWORK	Sees only opaque commitments. It cannot read amounts, parties, or balances, nor link one transfer to the next.	PROVEN
THE SENDER	Cannot watch where or when the recipient later spends what was sent.	PROVEN
ANYONE WITH THE ADDRESS	Sees one published key; every sender derives a fresh, indistinguishable transaction key.	PROVEN

ACT III · THE ANSWER

NO BRIDGE, NOTHING TO HACK.

The largest loss category in digital finance — designed out, not patched.

Bridges concentrate risk: they hold value and relay a forgeable message. Unicity holds nothing — a locked source-chain asset becomes a self-contained token the recipient verifies directly against the source contract.



ACT III · THE ANSWER

SETTLEMENT WITH NO INTERMEDIARY.

Trustless atomic swaps between self-contained bearer assets.

The old fix is a timed lock, and the clock is the attack surface — one side stalls and walks. Unicity removes it with **Predicate Swaps**: both commit independently, and the swap forms for both at once or not at all.

HASHED TIMELOCK (HTLC)

1 · A locks, reveals hash y

2 · B locks against same y

3 · A claims, revealing preimage x

4 · B must claim before timeout

THE TIMING TRAP

B late on step 4 — a dropped connection — and A keeps both.

UNICITY PREDICATE SWAP

A COMMITS
locks their token

B COMMITS
locks their token

UNICITY SERVICE
the shared reference

AUTOMATIC FAIRNESS

✓ both locked → swap completes
✓ either walked away → both refunded
✓ no deadlines, no chasing, done

ACT III · THE ANSWER

TWELVE STEPS BECOME FIVE.

The same friction operators spend billions trying to solve — gone.

Today x402 still routes settlement through a facilitator and a public chain. Unicity embeds authorization and proof in the token and cuts the handshake from **twelve steps to five**.

TRADITIONAL X402 — 12 STEPS

1 Client requests the resource
2 Server returns 402 Payment Required
3 Client signs and submits payment
4 ~~Server forwards to facilitator~~
5 ~~Facilitator verifies on-chain balance~~
6 ~~Settle on the shared ledger~~
7 ~~Await block confirmation~~
8 ~~Facilitator confirms to server~~
… then deliver the resource

UNICITY X402 — 5 STEPS

1 Client requests the resource
2 Server returns 402 Payment Required
3 Client pays with a bearer token (auth + proof inside)
4 Server verifies the token locally
5 Server delivers the resource

7 STEPS ELIMINATED

No facilitator. No shared ledger. No block wait.

ACT III · THE ANSWER

A MARKET WITH NO VENUE.

CEX speed and DEX custody, with native privacy and compliance — at once.

Legacy venues force a compromise. Unicity delivers exchange speed, self-custody, privacy, and compliance — with **no venue at all**.

	LEGACY CEX	LEGACY DEX	UNICITY
SPEED	sub-second	block time	sub-second
CUSTODY	custodial	self-custodial	self-custodial
PRIVACY	operator sees all	fully public	unlinkable
COMPLIANCE	off-chain	none	in the asset

ACT III · THE ANSWER

AND A WHOLE COMPANY RUN BY AGENTS.

Agents become the new smart contracts, executing verifiable logic directly on bearer assets.

Unicity built a decentralized autonomous corporation for BlackRock — a parametric insurer whose capital, underwriting, and cession all run as agents. Settlement is only the entry point: **the same protocol runs the whole corporation**.

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ACT IV · THE LINEAGE

BUILT BY INFRASTRUCTURE VETERANS.

The cryptography behind Unicity has run at nation scale since 2012.

GUARDTIME / KSI — DEPLOYED WITH

NATO OTAN · DARPA · LOCKHEED MARTIN · verizon · BIS · MAERSK · esa European Space Agency

THE TEAM

Founders built Guardtime's **KSI** — sovereign-grade verification, in production since 2012, eIDAS-grade. That lineage sustained **300,000 tx/sec** in Eesti Pank's 2021 test (heritage, not a live Unicity figure).

THE PROOF

Three mathematical papers prove the **privacy** and **no-double-spend** guarantees — public on github.com/unicitynetwork. Verifiable today, not asserted.

THE ALIGNMENT

Identity, speed, and peer-to-peer settlement in one instrument — the fair-access infrastructure Hard Yaka has backed for a generation.

ACT IV · THE NEXT STEP

SHIP THE FIRST COMPLIANT DOLLAR.

A regulated dollar that settles peer-to-peer the moment the Receive Predicate is satisfied.

Partner with a regulated issuer to bind identity, speed, and settlement into one instrument.

The first asset where the proof of who you are — and what you may do — travels with the money itself.

Institutions rely on walled gardens for compliance. Unicity embeds that compliance directly into the dollar.

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